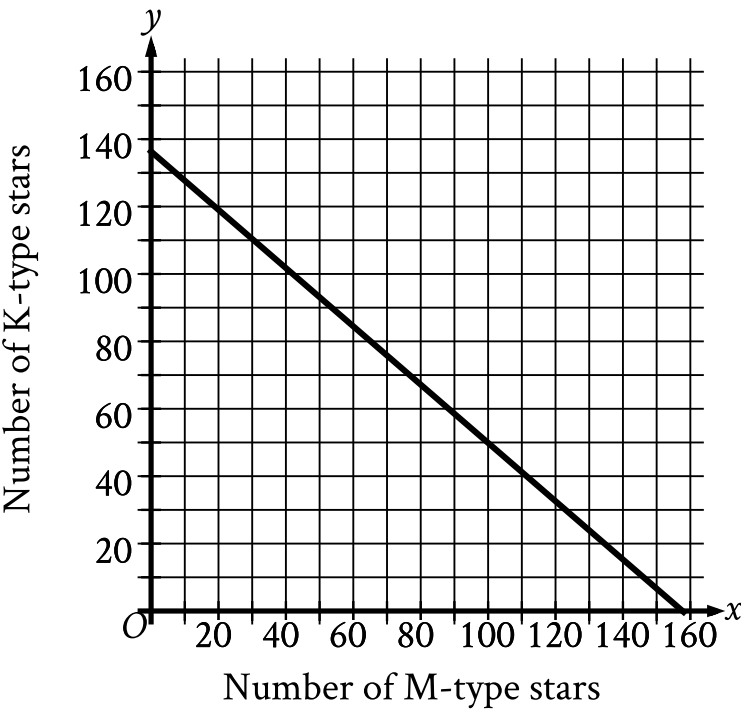


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question



A certain open star cluster contains M-type stars and K-type stars. The estimated total mass of M-type and K-type stars in this open star cluster is **127,882 quettagrams**. The graph shown models the possible combinations of the number of M-type stars, x , and K-type stars, y , that could be in this open star cluster if all the M-type stars have the same estimated mass and all the K-type stars have the same estimated mass. Based on the graph, which of the following is closest to the estimated mass, in quettagrams, of each M-type star in this cluster?

- Answer**
- A. 811
 - B. 938
 - C. 51,904
 - D. 75,978

Correct Answer: A

Rationale

Choice A is correct. The x-intercept of the graph shown represents the situation in which the cluster contains only M-type stars. Based on the graph, the x-intercept is approximately $(158, 0)$. Therefore, if there are **0** K-type stars in the cluster, then there are approximately **158** M-type stars in the cluster. It's given that the total mass of the stars in this cluster is **127,882** quettagrams. Therefore, the estimated mass of each M-type star is approximately $\frac{127,882}{158}$ quettagrams, or approximately **809.38** quettagrams. Thus, of the given choices, **811** is the closest to the estimated mass, in quettagrams, of each M-type star in this cluster.

Choice B is incorrect. This is the value closest to the estimated mass, in quettagrams, of each K-type star, not each M-type star, in this cluster.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.